

### Worksheet (WS) 3 – Partitioning Around Medoids (PAM) – how it works

- (a) If your data contains outliers, which distance - Manhattan distance (L1 Norm) or Euclidean distance (L2 Norm) – should give more robust results?
- (b) If your data contains outliers, which partitioning method – k-means or k-medoids – should give more robust results?
- (c) Is the PAM algorithm a k-means or k-medoids clustering method? What does PAM stand for?

Suppose we measure two variables  $X_1$  and  $X_2$  for each of four items A, B, C and D. The data are given in the following table

| Object | Observations |       |
|--------|--------------|-------|
|        | $x_1$        | $x_2$ |
| A      | 5            | 3     |
| B      | -1           | 1     |
| C      | -4           | 4     |
| D      | -3           | -2    |

Our intention in this exercise is to perform PAM clustering completely by hand to group the above dataset into two groups ( $k=2$ ).

The creators of PAM (Kaufman and Rousseeuw) in their 1987 document *Clustering by means of medoids* break down the PAM procedure into two stages – a BUILD stage and a SWAP stage.

Link: <https://wis.kuleuven.be/stat/robust/papers/publications-1987/kaufmanrousseeuw-clusteringbymedoids-l1norm-1987.pdf>

For your convenience, the relevant sections from that paper describing the BUILD and SWAP phases are extracted and shown in the textboxes below:

The algorithm we are using in PAM consists of two phases. In a first phase, called BUILD, an initial clustering is obtained by the successive selection of representative objects until  $k$  objects have been found. The first object is the one for which the sum of the dissimilarities to all other objects is as small as possible. This object is the most centrally located in the set of objects. Subsequently, at each iteration, another object is selected which decreases the sum (over all objects) of the dissimilarities to the most similar selected object, as much as possible. The process is continued until  $k$  objects have been found.

In the second phase of the algorithm (called SWAP), it is attempted to improve the set of representative objects and therefore also to improve the clustering yielded by this set. This is done by considering all pairs of objects  $(i, h)$  for which object  $i$  has been selected and object  $h$  has not. It is determined what effect is obtained when a swap is carried out; i.e. when object  $i$  is no longer selected as a representative object but object  $h$  is. The most profitable pair is then indeed swapped. This procedure is iterated until no further reduction is possible. As all potential swaps are considered, the results of the algorithm do not depend upon the order of the objects in the input file (except in case some of the distances between objects are tied). For more details see Kaufman and Rousseeuw (1987).

(d) Using Manhattan distance, create a dissimilarity matrix showing the distance between all pairs of objects.

(e) The BUILD stage aims to create an initial clustering with  $k$  medoids. In our case,  $k = 2$ . Show how the first medoid is selected.

(f) After the first medoid is selected, show how the second medoid is selected.

- (g) With the  $k=2$  medoids selected, assign the non-medoids to the respective clusters. This completes the BUILD's objective of obtaining an initial cluster. Before proceeding with the SWAP stage to see if this initial clustering can be improved, calculate the Cost of this initial cluster. Cost = sum of all dissimilarities from a non-medoid object to the nearest medoid.
- (h) We now proceed to the SWAP stage to see if we can find a better clustering result with a lower cost. If you study the algorithm carefully (see textbox), we can infer that one has to do  $k(n-k)$  cost calculations to complete one iteration of the SWAP phase to determine if there is a better clustering result compared to the initial grouping produced by the BUILD phase.  $n$  = total number of objects (4 in our case) and  $k$  = number of clusters (2 in our case). Explain how the formula  $k \times (n-k)$  is derived. Using  $n = 4$  and  $k = 2$ , how many cost calculations do we need to complete for one iteration of the SWAP phase?
- (i) If we use the notation  $C_{AB}$  to mean the cost when A and B objects are serving as medoids, write down all  $k(n-k)$  cost calculations that need to be performed (no need to calculate costs yet).
- (j) Perform all 4 cost calculations one at a time. (This part could be tedious and need lots of writing space).

- (k) Is there any cost from the SWAP stage that is lower than the cost calculated from the BUILD stage? If yes, do the swapping of medoids and repeat the SWAP procedure using the new set of medoids. If no, your answer from the BUILD stage is the final clustering results (which might not be the optimal global solution).
- (l) The PAM algorithm typically yields a good but not necessarily optimal solution. Can you see why? Hint: are all pairs (since  $k=2$ ) of objects considered in the BUILD/SWAP phases? Which pair(s) were not considered in our exercise?
- (m) Do you need the original data values of all 4 objects in order to complete the PAM algorithm or is it sufficient simply to use the dissimilarity matrix constructed in part (d)? This helps you understand why you can provide PAM with the distance matrix produced by `daisy()` to perform clustering when you have mixed data.